OPENER

Opener: an Open NR+ stack together

Axel Moinet · Codium Electronique

DECT World 2026 · Pre-event hands-on · 2 November 2026

Opener Initiative

History

Jan 2026

Kick-off: academia, vendors, device makers

Mar

Decision: **Zephyr** and **C**

Jun

Zephyr module bootstrapped

Sep

First driver merged

Goals

- **Academia & standardisation**: prototype, measure, feed results back to ETSI
- **Conformance & interop**: one open reference to test against
- **Integrated by vendors**: Apache-2.0, express patent grant
- **Lower the entry cost**: no need to write a full stack first

Technical choices: why Zephyr and C?

Zephyr

- Official base of the **Nordic** and **LMS** SDKs
- Well known and widely used in industry
- Net stack, `net_buf`, Kconfig, devicetree, Twister: ready to use
- Opener is an **out-of-tree module** developed against vanilla Zephyr

C

- The language of Zephyr and of the vendor SDKs
- What embedded teams already write, review and certify
- Coding rules follow Zephyr's guidelines (MISRA-based), checked by `clang-format` in CI

Zephyr-first, not Zephyr-only

Keep the boundary small

- The stack core relies on a **narrow set of primitives**: mutexes, locks, work queues, network buffers
- The contract is the **Zephyr API** itself
- Another RTOS or Linux only needs a thin platform shim implementing that subset
- The hardware is reached only through the **generic driver API**

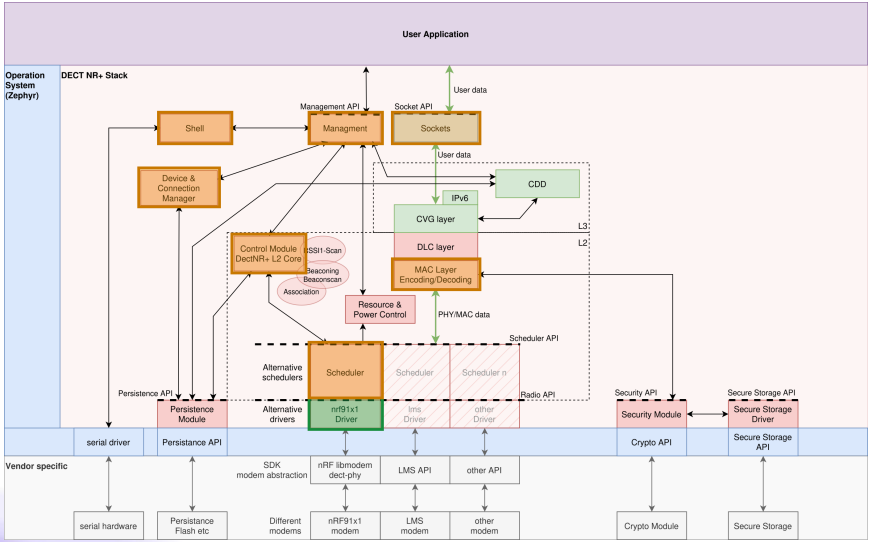
Today the code is Zephyr-only; making this boundary explicit is ongoing work.

First interop peer: hophop

- Independent NR+ stack written in **Rust** (Ariel OS)
- Async, heapless, IPv6 / CoAP oriented
- A second, independent implementation is the best interop test Opener can get

Architecture

done: driver API + nRF91x1 driver **next:** scheduler, L2 core, MAC, management, shell, sockets



Architecture proposal by deveritec

Roadmap: bottom-up

✔ Milestone 1	Driver API, nRF91x1 driver: RSSI1, RX, TX. Ping-pong and sniffer samples
→ Milestone 2	L2 in the Zephyr net stack, simple scheduler, transparent MAC, raw socket, management API + shell
Milestone 3	MAC encoding of beacons and association IEs. FT beaconing, PT sync, association
Beyond	Spinel host interface, HARQ, DLC and CVG services, security, power control, more radios (LMS)

Based on the deveritec roadmap proposal; ordering is discussed openly in the bi-weekly meetings.

Today’s hands-on: three tables

Codium Electronique

NR+Gate & NR+Node
Opener running on product hardware

deveritec

dectnrPi
Raspberry Pi shield & NR+ sniffer

hophop
Rust · Ariel OS

NR+ scanner
A second, independent stack

Codium: NR+Gate & NR+Node

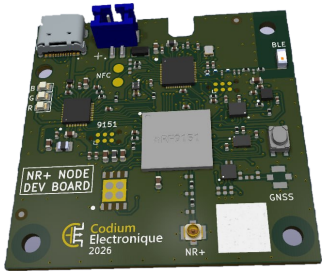
Hardware

- **NR+Gate**: nRF9151 + Linux module, PoE, Ethernet, Wi-Fi, LTE
- **NR+Node**: nRF9151 + nRF54L15 (BLE), sensors, battery

On the table

- **Opener ping-pong** between gateway and node
- **NR+ sniffer with web UI** on the gateway, via a PoC **Spinel** driver mode
- **Sensor demo**: node sends temperature, humidity, accelerometer, magnetometer and battery data





deveritec: dectnrPi

Hardware

- Raspberry Pi expansion board, or standalone over USB-C
- nRF9151 today; LMS-based pin-compatible variant in development
- SIM_DET and COEX1 on pin headers: **synchronise multiple modems**

On the table

- **Opener NR+ sniffer**: live NR+ traffic decoded in **Wireshark**
- Authors of the driver API and ping-pong sample
- Custom demos



hophop: NR+ in Rust

The stack

- NR+ stack written in **Rust**, on **Ariel OS**
- Async and heapless; IPv6 and CoAP oriented
- Runs on the same nRF9151 hardware

On the table

- **NR+ scanner**: what is on the air, per channel
- **CoAP servers** and **PC clients** joining dect_shell's network
- First stack lined up to **interoperate** with Opener

Why it matters

Two independent implementations of the same standard, in two languages, on two operating systems.

That is how interop bugs and spec ambiguities surface, and where we bring them back to ETSI.

Join us

11

TSG members

10+

organisations

21

bi-weekly meetings

Code	github.com/Opener-Initiative Apache-2.0, DCO sign-off
Website	opener-initiative.org
Chat	Matrix: `#opener-initiative:matrix.org`
Meetings	Bi-weekly, Wednesdays 14:00 CET: just ask to join